

Spring Data JPA - Quick Example

Student.java

```
package com.cognizant.spring_data_jpa_handson.entity;
```

```
import jakarta.persistence.Entity;  
import jakarta.persistence.GeneratedValue;  
import jakarta.persistence.GenerationType;  
import jakarta.persistence.Id;
```

```
@Entity
```

```
public class Student {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private int id;
```

```
    private String name;
```

```
    private String email;
```

```
    public Student() {  
    }
```

```
    public Student(String name, String email) {  
        this.name = name;  
        this.email = email;  
    }
```

```
    public int getId() {  
        return id;  
    }
```

```
    public String getName() {  
        return name;  
    }
```

```
    public String getEmail() {  
        return email;  
    }
```

```

    public void setId(int id) {
        this.id = id;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setEmail(String email) {
        this.email = email;
    }
}

```

STUDENTREPOSITORY.JAVA

```

package com.cognizant.spring_data_jpa_handson.repository;

import org.springframework.data.jpa.repository.JpaRepository;

import com.cognizant.spring_data_jpa_handson.entity.Student;

public interface StudentRepository
    extends JpaRepository<Student, Integer> {

}

```

SPRINGDATAJPAHANDSONAPLLICATION.JAVA:

```

package com.cognizant.spring_data_jpa_handson;

import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

import com.cognizant.spring_data_jpa_handson.entity.Student;
import com.cognizant.spring_data_jpa_handson.repository.StudentRepository;

@SpringBootApplication
public class SpringDataJpaHandsonApplication
    implements CommandLineRunner {

    private final StudentRepository studentRepository;

```

```

public SpringDataJpaHandsonApplication(
    StudentRepository studentRepository) {

    this.studentRepository = studentRepository;
}

public static void main(String[] args) {

    SpringApplication.run(
        SpringDataJpaHandsonApplication.class,
        args);
}

@Override
public void run(String... args) {

    Student student =
        new Student(
            "Thanisha",
            "thanisha29@gmail.com");

    studentRepository.save(student);

    System.out.println(
        "Student saved successfully");
}
}

```

APPLICATION PROPERTIES :

```

spring.application.name=spring-data-jpa-handson
spring.datasource.url=jdbc:mysql://localhost:3306/college
spring.datasource.username=root
spring.datasource.password=Tha*****
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true

```

OUTPUT :

```
@ Javadoc Console Progress History
terminated SpringDataJpaHandsonApplication [Java Application] C:\Users\Thanisha G\p2\pool\plugins\org.eclipse.just.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (27-Jul-2026, 10:28:06 pm - 10:28:13 pm elapsed 0:00:07) [pid: 28804]

:: Spring Boot :: (v4.1.0)

2026-07-27T22:28:08.143+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] c.c.s.SpringDataJpaHandsonApplication : Starting SpringDataJpaHandsonApplication using Java 21.0.11 with PID 28
2026-07-27T22:28:08.148+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] c.c.s.SpringDataJpaHandsonApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-27T22:28:08.887+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-07-27T22:28:08.954+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 55 ms. Found 1 JPA reposi
2026-07-27T22:28:09.386+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] org.hibernate.orm.jpa : HHH000540: Processing PersistenceUnitInfo [name: default]
2026-07-27T22:28:09.479+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.4.1.Final
2026-07-27T22:28:10.081+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-07-27T22:28:10.121+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-27T22:28:10.918+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@4eeb14
2026-07-27T22:28:10.923+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-07-27T22:28:11.226+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: college/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-07-27T22:28:12.646+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction.jta.pl
2026-07-27T22:28:12.740+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-07-27T22:28:12.858+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-07-27T22:28:13.304+05:30 INFO 28804 --- [spring-data-jpa-handson] [main] c.c.s.SpringDataJpaHandsonApplication : Started SpringDataJpaHandsonApplication in 3.617 seconds (process runni
Hibernate: insert into student (email,name) values (?,?)
Student saved successfully
2026-07-27T22:28:13.528+05:30 INFO 28804 --- [spring-data-jpa-handson] [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'default'
2026-07-27T22:28:13.532+05:30 INFO 28804 --- [spring-data-jpa-handson] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-07-27T22:28:13.546+05:30 INFO 28804 --- [spring-data-jpa-handson] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
```

Difference between JPA, Hibernate and Spring Data JPA

STUDENT.JAVA

```
package com.cognizant.jpa_hibernate_springdata_difference.entity;
```

```
import jakarta.persistence.Entity;
```

```
import jakarta.persistence.GeneratedValue;
```

```
import jakarta.persistence.GenerationType;
```

```
import jakarta.persistence.Id;
```

```
@Entity
```

```
public class Student {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private int id;
```

```
    private String name;
```

```
    private String email;
```

```
    public Student() {
```

```
    }
```

```

public Student(String name, String email) {
    this.name = name;
    this.email = email;
}

public int getId() {
    return id;
}

public String getName() {
    return name;
}

public String getEmail() {
    return email;
}
}

```

STUDENTREPOSITORY.JAVA

```

package com.cognizant.jpa_hibernate_springdata_difference.repository;

import org.springframework.data.jpa.repository.JpaRepository;

import com.cognizant.jpa_hibernate_springdata_difference.entity.Student;

public interface StudentRepository
    extends JpaRepository<Student, Integer> {

}

```

JPASTUDENTSERVICE.JAVA

```

package com.cognizant.jpa_hibernate_springdata_difference.service;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

import com.cognizant.jpa_hibernate_springdata_difference.entity.Student;

import jakarta.persistence.EntityManager;
import jakarta.persistence.PersistenceContext;

@Service

```

```

public class JpaStudentService {

    @PersistenceContext
    private EntityManager entityManager;

    @Transactional
    public void saveStudent(Student student) {

        entityManager.persist(student);

        System.out.println(
            "Student saved using JPA EntityManager");
    }
}

```

JPAHIBERNATESPRINGDATADIFFERENCEAPPLICATION.JAVA

```

package com.cognizant.jpa_hibernate_springdata_difference;

import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

import com.cognizant.jpa_hibernate_springdata_difference.entity.Student;
import com.cognizant.jpa_hibernate_springdata_difference.repository.StudentRepository;
import com.cognizant.jpa_hibernate_springdata_difference.service.JpaStudentService;

@SpringBootApplication
public class JpaHibernateSpringdataDifferenceApplication
    implements CommandLineRunner {

    private final JpaStudentService jpaStudentService;
    private final StudentRepository studentRepository;

    public JpaHibernateSpringdataDifferenceApplication(
        JpaStudentService jpaStudentService,
        StudentRepository studentRepository) {

        this.jpaStudentService = jpaStudentService;
        this.studentRepository = studentRepository;
    }

    public static void main(String[] args) {

```

```

SpringApplication.run(
    JpaHibernateSpringdataDifferenceApplication.class,
    args);
}

```

@Override

```

public void run(String... args) {

```

```

// JPA APPROACH

```

```

Student student1 =
    new Student(
        "Dharshini",
        "dharshini28@gmail.com");

```

```

jpaStudentService.saveStudent(student1);

```

```

// SPRING DATA JPA APPROACH

```

```

Student student2 =
    new Student(
        "Ravi",
        "ravi@gmail.com");

```

```

studentRepository.save(student2);

```

```

System.out.println(
    "Student saved using Spring Data JPA");
}
}

```

OUTPUT :

```

=====
:: Spring Boot ::                (v4.1.0)

026-07-27T22:32:20.048+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:20.054+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:20.869+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:20.937+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:21.402+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:21.457+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:22.074+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:22.132+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:22.843+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:22.847+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:23.012+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:25.046+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
Database JDBC URL [jdbc:mysql://localhost:3306/college]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: college/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
026-07-27T22:32:24.393+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:24.507+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:24.737+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:25.046+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
hibernate: insert into student (email,name) values (?,?)
student saved using JPA EntityManager
hibernate: insert into student (email,name) values (?,?)
student saved using Spring Data JPA
026-07-27T22:32:25.284+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:25.293+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
026-07-27T22:32:25.323+05:30 INFO 22876 --- [jpa-hibernate-springdata-difference] [
main] HibernateSpringdataDifferenceApplication : Starting JpaHibernateSpringdataDifferenceApplication using
main] HibernateSpringdataDifferenceApplication : No active profile set, falling back to 1 default profile: ""
main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 59 ms. Found 1
main] org.hibernate.orm.jpa : HHH0008540: Processing PersistenceUnitInfo [name: default]
main] org.hibernate.orm.core : HHH000000: Hibernate ORM core version 7.4.1.Final
main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.Connection
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:

main] org.hibernate.orm.core : HHH000409: No JTA platform available (set 'hibernate.transa
main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit '
main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will b
main] HibernateSpringdataDifferenceApplication : Started JpaHibernateSpringdataDifferenceApplication in 5.66

```

